William D. Legault

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EDUCATION

Northeastern University **Boston, MA**

Bachelor of Science in Computer Science and Biology *Expected Graduation: December 2027*

Coursework: Advanced Genomics, Biochemistry, Biostatistics (R), Genetics & Molecular Biology, Algorithms & Data Structures, Database Design, Object-Oriented Design & Cloud Computing

TECHNICAL SKILLS

Bioinformatics & Data: R, Python, Biopython (sequence alignment), ClinVar API, statistical analysis in R, data visualization
Wet Lab: PCR, gel electrophoresis, SDS-PAGE, DNA/RNA quantification (NanoDrop), cell culture
Software: TypeScript, JavaScript, Java, Kotlin; React, React Native, Next.js, Flask, Express; PostgreSQL, MongoDB, Supabase, AWS (Lambda, S3)

BIOINFORMATICS PROJECT

Genomic Variant Analyzer | *Python Flask, Biopython, ClinVar API, Next.js, TypeScript, Supabase, Recharts*

- Built a full-stack DNA variant analysis tool with a Python Flask backend that uses Biopython's PairwiseAligner to align sequences and classify coding consequences, including missense, nonsense, and frameshift variants.
- Integrated the ClinVar API to retrieve pathogenicity data for each identified variant, linking computational classification to clinical significance.
- Generated plain-language variant explanations and interactive Recharts visualizations in a Next.js frontend, making genomic results interpretable for non-specialist users.

EXPERIENCE

Sculpt.ai **Remote (New York, NY)**

Software Engineer (Part-Time) *Oct. 2025 – Present*

- Built and shipped React-based analytics tools exposing strength prediction (1RM) and nutrition planning (BMR/TDEE) algorithms to health professionals via REST APIs so they can verify accuracy.
- Merged 165+ pull requests across 6 repositories as a full-stack engineer on a React Native/Expo fitness app (iOS & Android, 1,000+ users), its GraphQL API (Node.js, Pothos, Prisma, PostgreSQL), and Next.js admin tools, using Apollo Client and React Context.
- Designed backend services for weekly streaks (atomic, race-safe updates under connection pooling), goal tracking, starter-workout generation, and exercise replacements ranked by target muscle, with Jest test coverage.
- Built the Next.js admin CMS supporting the beta rollout (onboarding waves, access controls, subscription management, content publishing), including an AWS Lambda/S3 media pipeline for 4:5 video and image re-cropping without re-upload, duplicate-exercise linking, a muscle-group hierarchy, and AI-suggested starting weights.

Northeastern University **Boston, MA**

Recreation Software Developer (Co-op) *July 2025 – June 2026*

- Led full-stack development as part of a 2-dev team on a React Native fitness app and admin CMS built to support a rollout to ~6,000 students, transforming a prototype into a production platform deployed via AWS and Vercel.
- Collaborated with managers to design admin workflows, implement live notifications, and streamline AWS infrastructure by optimizing database interactions and load balancer usage for performance and scalability.
- Developed, tested, and optimized the app for iOS and Android, using crash reports and custom Notion API–based bug reports to improve reliability and user experience.

Northeastern College of Arts, Media and Design **Boston, MA**

Immersive Media Lab Office Assistant *Jan. 2025 – April 2025*

- Educated students on VR technology and creative platforms through 10+ workshops and presentations, supporting Unity, Blender, Adobe, and Rokoko motion capture, including equipment maintenance and troubleshooting.

ADDITIONAL PROJECTS

Sophia: Philosophical Journal | *React, Supabase, Node.js, Express*

- Built a mental health journaling web app with CRUD functionality, a mood analytics dashboard, full-text PostgreSQL search, email-based auth, and tagging to analyze patterns in user data.

ACTIVITIES

Hackathons (x2) | Volunteer Snow Removal for local churches & senior living | Lifeguard and CPR/AED Certified